

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 16

In this class we finish the proof that all PIDs are UFDs. We also look at an immediate, but non-trivial consequence of this fact.

1.1. PIDs are UFDs. We will now prove that every PID is a UFD. We recall the following important result from last class.

Lemma 1.1. *Let R be a commutative ring and let $I_1 \subset I_2 \subset I_3 \subset \dots$ be ideals in R , then:*

$$(1.1) \quad I_\infty = \bigcup_{i=1}^{\infty} I_i$$

is an ideal in R .

Proof. Note that $0 \in I_1 \subset I_\infty$. Let $a, b \in I_\infty$ then $a \in I_i$ and $b \in I_j$ so $a, b \in I_{\max(i,j)}$. Since $I_{\max(i,j)}$ is an ideal so $-a, a+b \in I_{\max(i,j)} \subset I_\infty$. It follows that $(I_\infty, +)$ is an abelian group. It remains to show that if $a \in I_\infty$ and $b \in R$ then $ab \in I_\infty$. Let $a \in I_\infty$ then $a \in I_i$ for some i . Since I_i is an ideal we have $ab \in I_i$ for every $b \in R$, so $ab \in I_i \subset I_\infty$ and we conclude that I_∞ is an ideal. ■

Lemma 1.2 (Ascending chain condition). *Let R be a PID. If $I_1 \subset I_2 \subset I_3 \subset \dots$ are nested ideals in R then there is some $j < \infty$ such that $I_{j+k} = I_j$ for all $k \geq 0$.*

Proof. Let I_∞ be defined as in the previous lemma. Since R is a PID we find some $a \in R$ such that $I_\infty = \langle a \rangle$. Since $a \in I_\infty$ there is some j such that $a \in I_j$ and we conclude that $I_\infty = \langle a \rangle \subset I_j \subset I_\infty$ so $I_j = I_\infty$. If $k \geq 0$ then $I_j \subset I_{j+k} \subset I_\infty = I_j$ which implies $I_{j+k} = I_j$. ■

Lemma 1.3. *Let R be a commutative ring. If $a \in R$ divides $b \in R$ then $\langle b \rangle \subset \langle a \rangle$. In particular, $\langle a \rangle = \langle b \rangle$ if and only if a and b are associate.*

Proof. If a divides b then $b = ac$, so we have:

$$(1.2) \quad \langle b \rangle = \langle ac \rangle = \{acr \mid r \in R\} = \{a(cr) \mid r \in R\} \subset \langle a \rangle$$

where we have used that any element $a(cr) \in \langle a \rangle$.

The last part of the lemma follows since if a and b are associate then a divides b and b divides a . Conversely, if $\langle a \rangle = \langle b \rangle$ then $a = bc$ and $b = ad$ from which it follows that $a = adc$, so if we cancel a on both sides (recalling that R is an integral domain), we

see that $1 = dc$ so d, c are units (and each others inverses). So we have $a = bc$ with $c \in R$ a unit, so a and b are associate. ■

Lemma 1.4. *Let R be a PID. Every $a \in R$ that is non-zero and not a unit is a product of irreducible elements.*

Proof. Let $a_0 \in R$ be non-zero and not a unit. If a_0 is irreducible, then there is nothing to prove. Assume instead that a_0 is reducible, and write $a_0 = a_1 b_1$ with a_1, b_1 both not units. Since a_1 divides a_0 , and a_0 and a_1 are not associate we have $\langle a_0 \rangle \subsetneq \langle a_1 \rangle$. If a_1 is not irreducible then we can continue and construct $a_1 = a_2 b_2$ with a_2, b_2 not units. We claim that eventually we must obtain a_j that is irreducible in this process. Indeed, suppose this was not the case, then we produce $\langle a_0 \rangle \subsetneq \langle a_1 \rangle \subsetneq \langle a_2 \rangle \subsetneq \dots$, which is an ascending sequence of ideals of R . But we know that any such sequence must terminate at some j : $\langle a_j \rangle = \langle a_{j+k} \rangle$ for $k \geq 0$. This is a contradiction, and we conclude that for some $j < \infty$ we have a_j is irreducible. We can now write:

$$(1.3) \quad a_0 = a_1 b_1 = a_2 b_1 b_2 = \dots = a_j b_1 b_2 \dots b_j$$

with a_j is irreducible. That is, a_0 is divided by at least one irreducible element. That is, for any $a \in R$, $a \neq 0$ and a not a unit, we can write $a = p_1 b_1$ with p_1 irreducible. If b_1 is irreducible we are done, otherwise we can write $b_1 = p_2 b_2$ with p_2 irreducible. Proceeding in this fashion we obtain $\langle a \rangle \subsetneq \langle b_1 \rangle \subsetneq \langle b_2 \rangle \subsetneq \dots$ which is an ascending sequence of ideals, and since this terminates we conclude that for some j we have that b_j must be irreducible and we obtain a as a product of irreducible elements. ■

The proof of the following two theorems are analogous the the corresponding results for polynomial rings, we only prove one of them.

Theorem 1.1. *Let R be a PID. An ideal $\langle p \rangle$ is maximal if and only if p is irreducible.*

Proof. Note that if $p \in R$ is not irreducible, then we can write $p = ab$, both of a, b not units, so a divides p and we obtain $\langle p \rangle \subsetneq \langle a \rangle$. On the other hand, note that a is not a unit so a is not associate to 1 which implies $\langle a \rangle \neq \langle 1 \rangle = R$. It follows that $\langle a \rangle$ is a proper ideal that properly contains $\langle p \rangle$, so $\langle p \rangle$ is not maximal. That is p not being irreducible implies that $\langle p \rangle$ is not maximal, and taking the contrapositive of this it follows that if $\langle p \rangle$ is maximal then p is irreducible.

Conversely, assume that p is irreducible. Assume that we have an ideal I such that $\langle p \rangle \subset I \subset R$, we claim that $I = \langle p \rangle$ or $I = R$, which proves that $\langle p \rangle$ is maximal. Since R is a PID we can write $I = \langle a \rangle$ for some $a \in R$, and since $p \in \langle p \rangle \subset I = \langle a \rangle$ we can write $p = ab$ for some $b \in R$. If b is a unit then a and p are associate so $\langle p \rangle = \langle a \rangle = I$. If a is a unit then a is associate to 1, so $R = \langle 1 \rangle = \langle a \rangle = I$. However, since p is irreducible and $p = ab$ we know that at least one of a and b is a unit and we conclude that either $I = \langle p \rangle$ or $I = R$. ■

Theorem 1.2. *Let R be a PID. If $p \in R$ is irreducible and p divides $ab \in R$ then p divides a or b .*

Proof. Note that $ab \in \langle p \rangle$ since p divides ab . Since $\langle p \rangle$ is a maximal ideal, the ideal $\langle p \rangle$ is also prime so if $a, b \in \langle p \rangle$ then $a \in \langle p \rangle$ or $b \in \langle p \rangle$. Assume that $a \in \langle p \rangle$ (otherwise rename a and b). Since $a \in \langle p \rangle$ we have $a = pc$ so p divides a . ■

The following corollary follows by induction:

Corollary 1.2.1. *Let R be a PID. If $p \in R$ is irreducible and p divides $a_1 \dots a_n \in R$ then p divides at least one factor a_j .*

Theorem 1.3. *Every PID, R , is a UFD.*

Proof. Let $a \in R$ be non-zero and not a unit. Let $a = p_1 \dots p_n$ be a decomposition of a into irreducible elements. Assume that we have some other $a = q_1 \dots q_m$ decomposition into irreducible elements. So we have $p_1 \dots p_n = q_1 \dots q_m$. Since p_1 is irreducible and divides $q_1 \dots q_m$ we know that p_1 divides one of the factors q_j , after reordering we may assume that p_1 divides q_1 . Since q_1 is also irreducible we have $q_1 = u_1 p_1$ for some unit u_1 so q_1 and p_1 are associates. Using that $q_1 = u_1 p_1$ we obtain $p_1 \dots p_n = p_1 u_1 q_2 \dots q_m$ after cancelling p_1 on both sides we have $p_2 \dots p_n = u_1 q_2 \dots q_m$. We can now use the same argument for p_2 ; p_2 must divide one of the factors $q_2, \dots, q_m$ and we assume after reordering that p_2 divides q_2 (note that p_2 can not divide u_1 since this would imply $u_1 = p_2 a$ or $1 = p_2 (a u_1^{-1})$ which would imply that p_2 is a unit which it is not). Assume now for simplicity that $n \leq m$ (if $m > n$ simply change the roles of p_j and q_j). We proceed by induction and find $u_1, \dots, u_n$ such that $q_j = u_j p_j$ for $j = 1, \dots, n$. If $m > n$ then we obtain after cancelling common factors an equality:

$$(1.4) \quad 1 = u_1 \dots u_n q_{n+1} \dots q_m = (u_1 \dots u_n q_{n+1} \dots q_{m-1}) q_m$$

from which it follows that q_m has a multiplicative inverse, and is therefore a unit. However, irreducible elements must not be units, so we conclude that $n = m$. That is, if $a = p_1 p_2 \dots p_n = q_1 q_2 \dots q_m$, with p_j, q_j irreducible, then $n = m$ and after rearranging we find units $u_1, \dots, u_n$ such that $q_j = u_j p_j$ which proves that the factorization is unique (in the sense of a UFD). ■

Theorem 1.4 (Fundamental theorem of arithmetic). *Any integer $n \in \{2, 3, 4, \dots\}$ has a unique prime decomposition.*

Proof. Since $\mathbb{Z}$ is a PID it is a UFD. Any $n \in \mathbb{Z}$ that is non-zero and not a unit can then be written, uniquely up to a unit, as a product of irreducible elements, which are plus and minus the primes in $\mathbb{N}$. Let $n \geq 2$ be an integer. We find primes $p_1, \dots, p_m$, uniquely determined up to ordering, and $\varepsilon_1, \dots, \varepsilon_m \in \{0, 1\}$ such that:

$$(1.5) \quad n = (-1)^{\varepsilon_1} p_1 \cdot (-1)^{\varepsilon_2} p_2 \cdot \dots \cdot (-1)^{\varepsilon_m} p_m = (-1)^{\varepsilon_1 + \dots + \varepsilon_m} p_1 \cdot \dots \cdot p_m,$$

and noting that the left-hand side is positive, we must have $(-1)^{\varepsilon_1 + \dots + \varepsilon_m} = 1$ and we conclude that:

$$(1.6) \quad n = p_1 p_2 \dots p_m.$$

That is, any $n \geq 2$ has a unique, up to reordering, decomposition into a product of primes. ■

Email address: sandfeldt@uchicago.edu